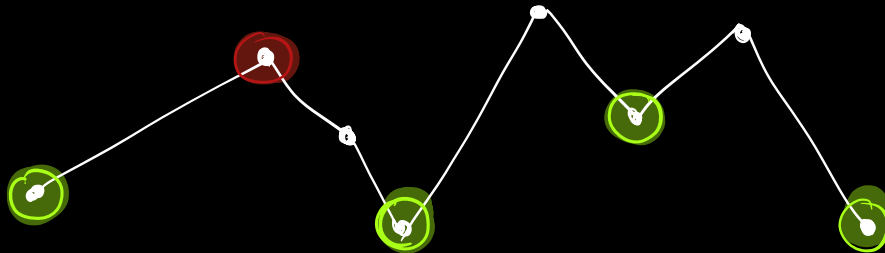


Q1. Given an unsorted array of N distinct elements, find any one of the local minima.

- ↳ smaller than adjacent elements:



will there always be a local minima

$\Rightarrow$ Yes

arr $\Rightarrow$ 9 8 7 3 6 4 1 5

arr $\Rightarrow$ 1 2 3 4

$$arr[i-1] > arr[i] < arr[i+1]$$

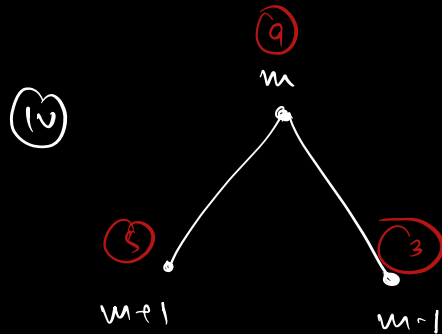
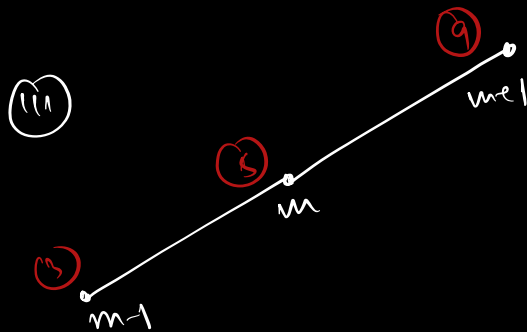
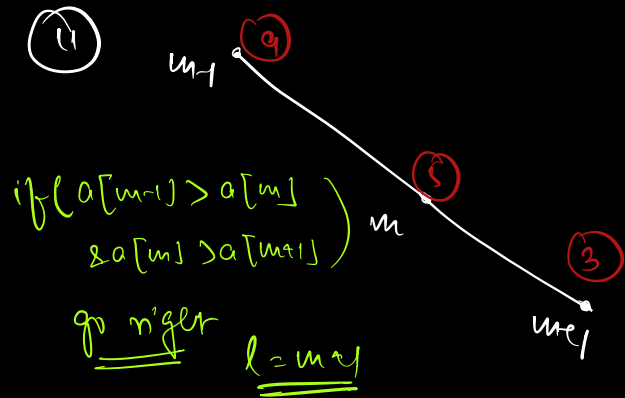
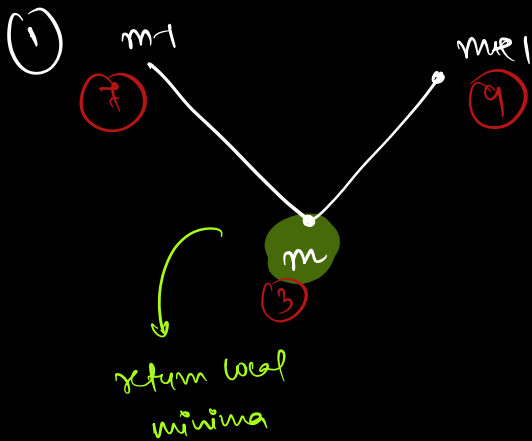
↑
local mining

mid
↓

$arr[mid] == k \rightarrow return$

$arr[mid] < k \rightarrow go\ right$

$arr[mid] > k \rightarrow go\ left$

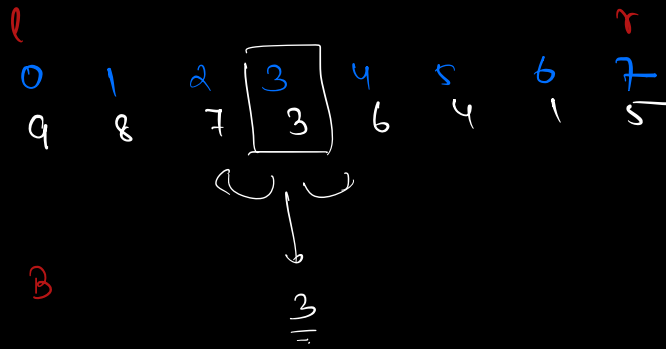


if ($a[m+1] > a[m]$ & $a[m] > a[m-1]$) {

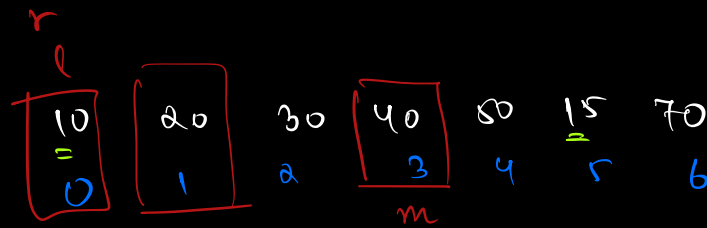
go left

$r = m-1$

go either
right or left.



$$m = \frac{0+7}{2} = 3$$



$$m = \frac{0+6}{2} = 3$$

return 10

$$r = m - 1 = 2$$

$$m = \frac{0+2}{2} = 1$$

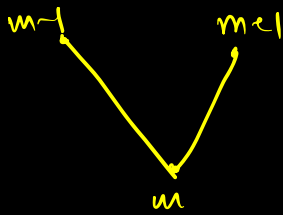
$$r = m - 1 = 0$$

$$m = \frac{0+0}{2} = 0$$

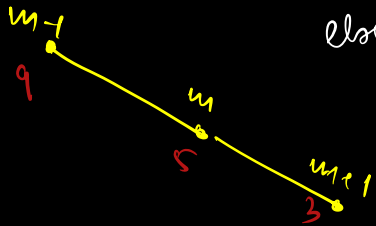
```

int minimum(arr) {
    if (n == 1) return arr[0]
    if (arr[0] < arr[1]) return arr[0]
    if (arr[n-1] < arr[n-2]) return arr[n-1]
    l = 1
    r = n - 2
    while (l <= r) {
        m =  $\frac{l+r}{2}$ 
    }
}

```



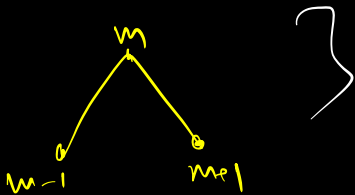
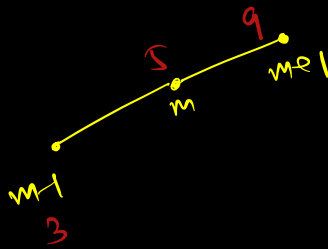
if ($arr[m] < arr[m+1]$
 $\&\& arr[m] < arr[m-1]$)
 return $arr[m]$



else if ($arr[m-1] > arr[m]$
 $\&\& arr[m] > arr[m+1]$) {
 $l = m+1$

else

$r = m-1$



$TC \Rightarrow O(\log N)$ $SC \Rightarrow O(1)$

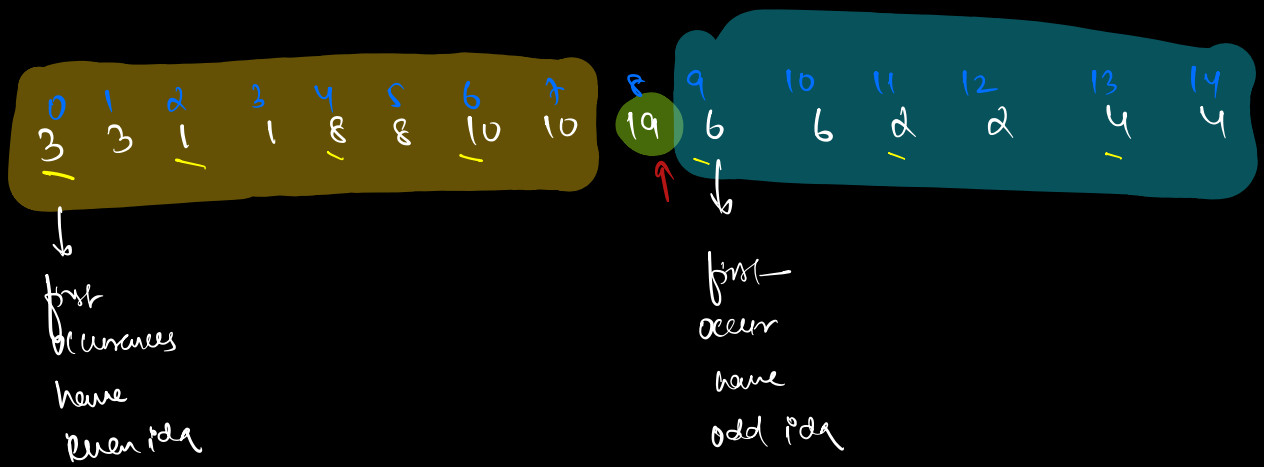
Q2. In arr, all elements are repeating twice except for one.

Find the unique element; NOTE $\rightarrow$ all duplicate elements are adjacent to each other

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
3	3	1	1	8	8	10	10	19	6	6	2	2	4	4

↑

$\Rightarrow \boxed{XOR \rightarrow O(N)}$



if (arr[m] != arr[m-1]
 && arr[m] != arr[m+1])

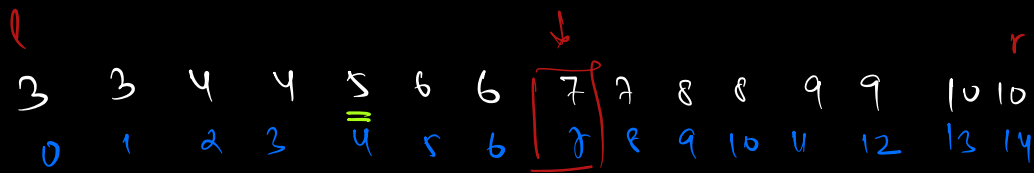
return arr[mid]

else if (1st occur is even idx)

l = m + 1

else

r = m - 1



$$m = \frac{0+14}{2} = 7$$

int unique(arr) {

[

while (l <= r) {

m = $\frac{l+r}{2}$

if (arr[m] != arr[m-1]
 && arr[m] != arr[m+1])

return arr[m];

first = m

if (arr[m-1] == arr[m])

first = m-1

if (first % 2 == 0) {

l = m+1

} else {

r = m-1

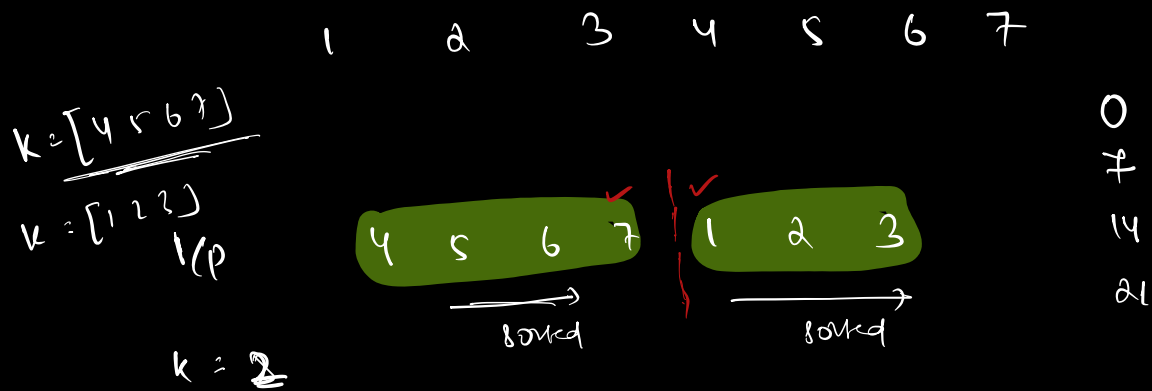
}

}

TC $\Rightarrow O(\log N)$

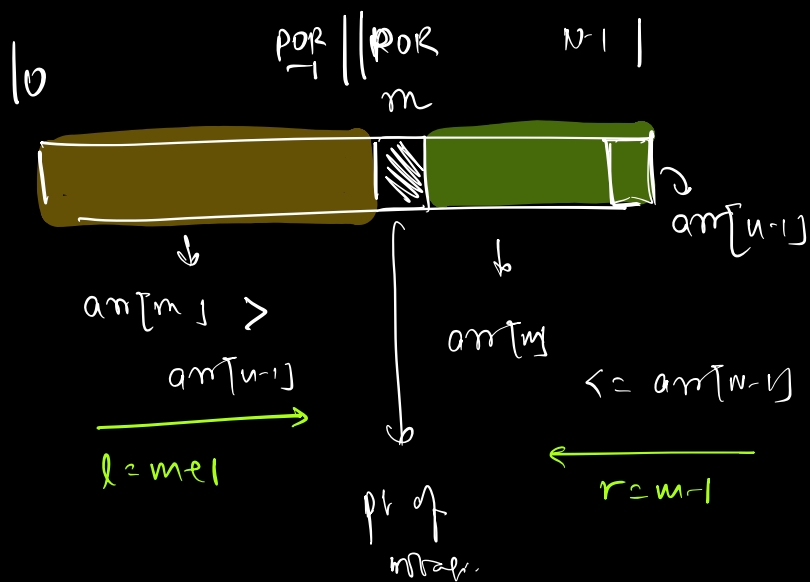
SC $\Rightarrow O(1)$

Q3. Given a ^{clockwise} rotated sorted array of distinct elements, find if k exists?



Basic $\rightarrow$ Linear search $\rightarrow O(N)$

- divided into 2 sorted arrays
- all elements (left) $>$ all elements (right)



$$arr[m-1] > arr[m] < arr[m+1]$$

$$l = 0$$

$$r = n-1$$

while($l \leq r$) {

$$m = \frac{l+r}{2}$$

if($arr[m-1] > arr[m] < arr[m+1]$) {

$$pos = m$$

break;

}

else if($arr[m] > arr[m+1]$) {

$$l = m+1$$

} else {

$$r = m-1$$

}

i) if $k > arr[N-1] \rightarrow$ left side

binarySearch(arr, l , 0, r , pos-1)

ii) if $k < arr[N-1] \rightarrow$ right side

binarySearch(arr, pos, l , r , N-1)

- i) finding POR (point of rotation) $\rightarrow \log N$
 ii) find which side is k
 iii) do BS on that side $\rightarrow \log N$

$$\begin{array}{l} \text{TC} \approx O(\log N) \\ \text{SC} \approx 1 \end{array}$$

Q4. Find sq. root of a eve no. Ch
 ↓
integer part

$$\begin{array}{l} 16 \rightarrow 4 \\ 52 \rightarrow 7 \\ 16 \rightarrow 4 \\ 70 \rightarrow 8 \end{array}$$

$$k = 70$$

$$1 \times 1 = 1$$

$$2 \times 2 = 4$$

$$3 \times 3 = 9$$

$$4^2 = 16$$

$$5^2 = 25$$

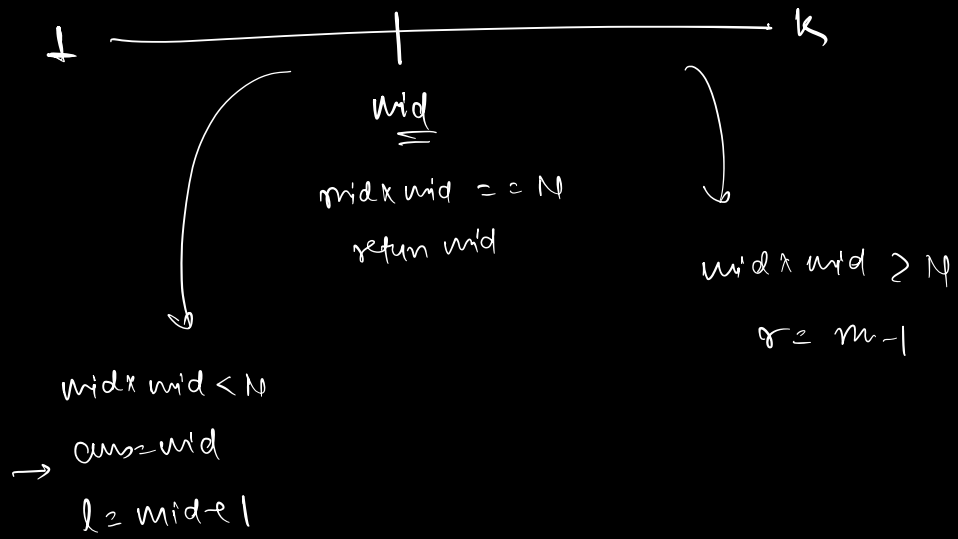
$$6^2 = 36$$

$$7^2 = 49$$

$$8^2 = 64$$

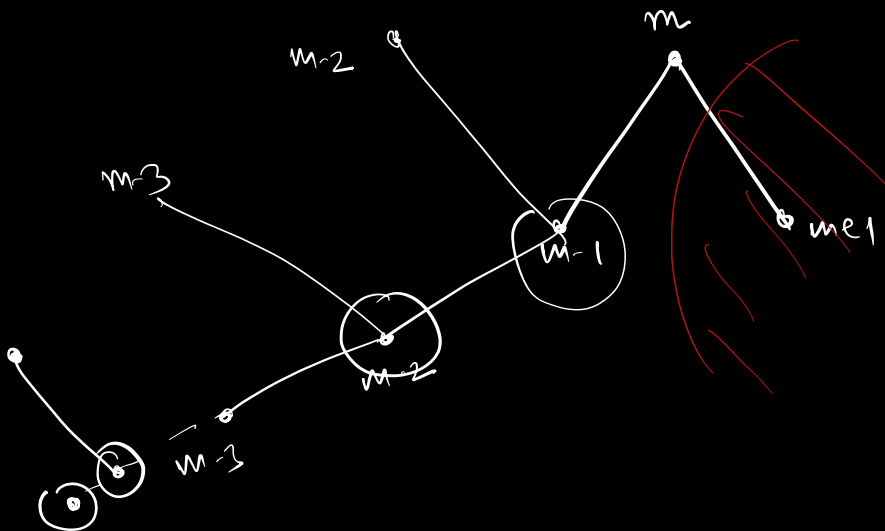
$$9^2 = 81 < 70$$

$$\text{TC} \approx \underline{\underline{O(\sqrt{N})}}$$



$return\ \max^m\ \underline{\underline{ans}}$

$TC \Rightarrow O(\log N)$
 $SC \Rightarrow O(1)$



—

9

13

151

子

10

8

